

## Lecture 26: Antiderivatives and Indefinite Integration

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**Goal:** Reverse the process of differentiation.

|                                                                        |                                                |
|------------------------------------------------------------------------|------------------------------------------------|
| $\int k f(x) \, dx = k F(x) + C$                                       | $\int (f(x) + g(x)) \, dx = F(x) + G(x) + C$   |
| $\int f(g(x)) g'(x) \, dx = F(g(x)) + C$                               | $\int (f(x) - g(x)) \, dx = F(x) - G(x) + C$   |
| $\int x^n \, dx = \frac{1}{n+1} x^{n+1} + C \text{ (when } n \neq -1)$ | $\int \frac{1}{x} \, dx = \ln  x  + C$         |
| $\int e^x \, dx = e^x + C$                                             | $\int e^{kx} \, dx = \frac{1}{k} e^{kx} + C$   |
| $\int \sin x \, dx = -\cos x + C$                                      | $\int \cos x \, dx = \sin x + C$               |
| $\int \sec^2 x \, dx = \tan x + C$                                     | $\int \sec x \tan x \, dx = \sec x + C$        |
| $\int \csc x \cot x \, dx = -\csc x + C$                               | $\int \csc^2 x \, dx = -\cot x + C$            |
| $\int \frac{1}{\sqrt{1-x^2}} \, dx = \sin^{-1} x + C$                  | $\int \frac{1}{1+x^2} \, dx = \tan^{-1} x + C$ |

Table 1: Table of Antiderivatives







